IIIT-H Exam: Questions & Explanations

My notes. Green = correct answer, red = what I picked when it was wrong. Q numbers follow the order of my screenshots.

Q1. What is the main objective of using Principal Component Analysis (PCA) in feature extraction?

A. To create new features that are combinations of original features, while maximizing the variance in the dataset. ✓

B. To reduce the number of features by selecting a subset that best represents the original data. (my pick)

C. To encode categorical variables into numerical features.

D. To cluster similar data points into groups.

I picked B, correct is A

Why: PCA does not pick some of your existing columns. It builds brand-new axes (principal components), each one a weighted combination of the original features. The first component points in the direction where the data varies the most, the second is the next-best direction at right angles to it, and so on. Keeping only the first few components gives fewer features while keeping most of the information (variance).

Others wrong because: B describes feature SELECTION (keeping a subset of the original columns unchanged). PCA creates new features, so B is wrong. C is encoding (e.g., one-hot encoding). D is clustering (e.g., K-Means).

Remember: PCA = new features made from combinations, chosen to maximize variance. Selecting a subset = feature selection, not PCA.

Note: This is the one question where the selected answer was wrong: the common trap is confusing extraction (new features) with selection (a subset of old features).

Q2. In scikit-learn, which method is used to train a model?

A. fit() ✓

B. transform()

C. predict()

D. score()

I picked A: correct

Why: In scikit-learn, calling model.fit(X_train, y_train) makes the model learn from the training data (for example, it computes the weights of a linear regression). After fitting, you use predict() to get outputs and score() to measure performance.

Others wrong because: transform() applies a learned transformation (e.g., a scaler or PCA) to data; it does not train a predictor. predict() produces outputs from an already-trained model. score() evaluates a trained model on data.

Remember: Train = fit(). Use = predict() / transform(). Evaluate = score().

Q3. In the context of machine learning, what does 'overfitting' refer to?

A. A model that is too simple to capture the underlying trend

B. A model that performs well on training data but poorly on new, unseen data ✓

C. A model that generalizes well to both training and unseen data

D. A model that cannot handle noisy data

I picked B: correct

Why: An overfitted model has memorized the training set, including its noise and quirks, instead of learning the real pattern. So training error is very low but error on new data is high.

Others wrong because: A describes underfitting (too simple). C describes a good model that generalizes well. D is not the definition: handling noise is a related symptom, but overfitting is defined by the train/test performance gap.

Remember: Overfit = great on train, bad on new data.

Q4. What is the primary goal of supervised learning?

A. To find patterns in unlabeled data

B. To predict the output for new, unseen data based on labeled training data ✓

C. To minimize the computational cost of algorithms

D. To explore the relationships between different data features

I picked B: correct

Why: In supervised learning every training example comes with a known answer (label). The model learns the mapping from inputs to outputs so it can predict the output for inputs it has never seen.

Others wrong because: A is unsupervised learning (no labels). C is an engineering concern, not the learning goal. D is exploratory data analysis, which is not the defining goal of supervised learning.

Remember: Supervised = labels in, predictions out.

Q5. What is the main difference between Standardization and Normalization in the context of data preprocessing?

A. Standardization scales data to a range between 0 and 1, while Normalization centers data around the mean.

B. Standardization transforms features to have a mean of 0 and a standard deviation of 1, while Normalization scales features to a specific range. ✓

C. Standardization is used for categorical data, whereas Normalization is for numerical data.

D. Standardization is only applicable for linear models, while Normalization is for non-linear models.

I picked B: correct

Why: Standardization (z-score) uses z = (x - mean) / std, so the result has mean 0 and standard deviation 1. Normalization (min-max) uses x' = (x - min) / (max - min), which squeezes values into a fixed range, usually [0, 1].

Others wrong because: A swaps the two definitions. C is wrong because both work on numerical features. D is wrong because both can be used with any model type; the choice depends on the data and the algorithm.

Remember: Standardize = z-score. Normalize = min-max range.

Q6. What is the purpose of normalizing data?

A. To convert data into a categorical format

B. To scale features so they have similar ranges and distributions ✓

C. To remove outliers from the data

D. To increase the number of features in the dataset

I picked B: correct

Why: Features often have very different scales (e.g., age 0-100 vs salary 0-1,000,000). Algorithms that use distances or gradients can be dominated by the large-scale feature. Normalizing puts features on comparable scales so each contributes fairly.

Others wrong because: A is binning/encoding, not normalization. C is wrong: scaling does not remove outliers (min-max can even be distorted by them). D is wrong: normalization does not add features.

Remember: Normalizing = comparable scales, not fewer outliers or more features.

Q7. What is the effect of applying the Z-score normalization to a dataset?

A. It ensures that all features have the same mean and variance ✓

B. It scales the data to a fixed range between 0 and 1

C. It removes categorical features from the dataset

D. It converts features into a logarithmic scale

I picked A: correct

Why: After z-score scaling every feature has mean 0 and variance 1 (standard deviation 1), so all features share the same mean and variance. The values are not limited to a fixed range.

Others wrong because: B describes min-max scaling. C is not something scaling does. D describes a log transform, a different technique.

Remember: Z-score -> mean 0, variance 1. Min-max -> [0, 1].

Q8. Which of the following statements accurately describes the MNIST dataset?

A. MNIST is a dataset consisting of images of handwritten digits (0-9) with each image being 64x64 pixels in size.

B. MNIST is a dataset used for natural language processing, containing text data from various sources.

C. MNIST is a dataset consisting of grayscale images of handwritten digits (0-9) with each image being 28x28 pixels in size. ✓

D. MNIST is a dataset used for time series analysis, containing sequences of stock market prices.

I picked C: correct

Why: MNIST has 70,000 grayscale images of handwritten digits 0-9 (60,000 training and 10,000 test), each 28 x 28 pixels, so each image flattens to a 784-length vector.

Others wrong because: A has the wrong image size (64x64). B and D describe NLP and time-series data, which MNIST is not.

Remember: MNIST = 28x28 grayscale digits.

Q9. When performing feature extraction using Autoencoders, what is the primary function of the encoder part of the network?

A. To reconstruct the input data from the compressed representation.

B. To learn a mapping from the original feature space to a lower-dimensional latent space. ✓

C. To apply a supervised learning algorithm to classify data.

D. To transform input features into a higher-dimensional space to improve classification performance.

I picked B: correct

Why: An autoencoder has two halves. The encoder compresses the input into a small latent vector (the extracted features). The decoder tries to rebuild the original input from that vector. The compressed latent code is what you use as the new, lower-dimensional features.

Others wrong because: A is the job of the decoder. C is wrong because autoencoders are trained in an unsupervised/self-supervised way. D is wrong because the usual goal is compression into fewer dimensions.

Remember: Encoder compresses, decoder reconstructs.

Q10. Which of the following methods can be used to detect underfitting?

A. Training and validation error both high ✓

B. Training error low, validation error high

C. Training error high, validation error low

D. Both errors are low

I picked A: correct

Why: An underfitted model is too simple to learn the pattern, so it performs badly even on the data it was trained on. Both training and validation errors are high.

Others wrong because: B is the signature of overfitting. C is very unusual (validation better than training can occur by chance but is not a standard indicator). D means the model is performing well.

Remember: Both high = underfit. Low train, high val = overfit.

Q11. In the context of unsupervised learning, what is clustering used for?

A. To predict a continuous output variable

B. To group similar data points together ✓

C. To classify data into predefined categories

D. To optimize model hyperparameters

I picked B: correct

Why: Clustering looks at unlabeled data and groups points so that points in the same cluster are more similar to each other than to points in other clusters (e.g., K-Means, DBSCAN).

Others wrong because: A is regression. C is classification (needs predefined labels). D is hyperparameter tuning, unrelated to clustering.

Remember: Clustering = group similar points, no labels.

Q12. What is the main goal of feature selection?

A. To increase the number of features

B. To reduce the dimensionality of the feature space ✓

C. To slow down the learning algorithm

D. To decrease the predictive accuracy

I picked B: correct

Why: Feature selection keeps only the most useful original features and drops the irrelevant or redundant ones. This reduces dimensionality, speeds up training, and often reduces overfitting.

Others wrong because: A, C and D are the opposite of the intended outcome of feature selection.

Remember: Selection = fewer original features.

Q13. In which scenario is a model most likely to overfit?

A. Large training dataset

B. High noise in data

C. Model has too many parameters ✓

D. Low model complexity

I picked C: correct

Why: A model with too many parameters has enough flexibility to memorize the training examples, including the noise, which leads to overfitting.

Others wrong because: A: more data usually REDUCES overfitting. B: noise can make overfitting worse, but it is the high capacity that actually lets the model memorize it, so C is the best answer. D: a low-complexity model tends to underfit.

Remember: Too many parameters = overfitting risk.

Q14. In PCA, the number of principal components is always:

A. Less than or equal to the number of input dimensions ✓

B. Greater than or equal to the number of input dimensions

C. Equal to the number of input dimensions

D. Depends on the variance threshold

I picked A: correct

Why: PCA can produce at most as many principal components as there are input features (in fact at most min(samples, features)). You usually keep fewer to reduce dimensionality, but you can never have more than the input dimensions.

Others wrong because: B is impossible. C is only true if you keep all components. D describes how many you CHOOSE to keep, not the upper bound that is always true.

Remember: PCs <= input dimensions, always.

Q15. Which of the following is not a common technique for feature extraction?

A. Principal Component Analysis (PCA)
B. Linear Discriminant Analysis (LDA)
C. t-Distributed Stochastic Neighbor Embedding (t-SNE)
D. K-Means Clustering ✓

I picked D: correct

Why: PCA, LDA and t-SNE all transform data into a new lower-dimensional representation. K-Means is a clustering algorithm that groups points; its main purpose is not to create a new feature space.

Others wrong because: A, B and C are all recognized dimensionality-reduction / feature-extraction techniques (t-SNE is mainly used for visualization).

Remember: K-Means groups points; it is not feature extraction.

Q16. Which of the following is not a common feature extraction technique?

A. Principal Component Analysis (PCA)
B. Linear Discriminant Analysis (LDA)
C. Backward feature selection ✓
D. Kernel PCA

I picked C: correct

Why: Backward feature selection starts with all features and removes the least useful ones one by one. It selects a subset of existing features and never creates new ones, so it is feature selection rather than extraction.

Others wrong because: PCA, LDA and Kernel PCA all build new features from combinations of the original ones.

Remember: Backward elimination = selection, not extraction.

Q17. What is the main benefit of feature extraction?

A. Increases the size of the dataset
B. Helps in reducing overfitting
C. Reduces the complexity of data ✓
D. Avoids normalization

I picked C: correct

Why: Feature extraction compresses many features into fewer, more informative ones, which reduces the complexity of the data (fewer dimensions, faster training, easier visualization).

Others wrong because: A is wrong: it does not add data. B is a possible side-effect but not the main benefit (and not guaranteed). D is wrong: extracted features often still need scaling.

Remember: Extraction -> simpler, lower-dimensional data.

Note: B is also a commonly cited secondary benefit. C is the best fit for 'main benefit'.

Q18. What is the purpose of cross-fold validation?

A. To reduce dimensionality

B. To estimate model performance ✓

C. To remove outliers

D. To avoid overfitting

I picked B: correct

Why: Cross-validation splits the data into k folds, trains on k-1 folds and tests on the remaining fold, repeating k times. Averaging the k scores gives a more reliable estimate of how the model will perform on unseen data.

Others wrong because: A and C are unrelated to cross-validation. D is a common misconception: cross-validation helps you DETECT overfitting and choose better settings, but it does not by itself prevent it.

Remember: Cross-validation estimates performance (it detects overfitting, it does not prevent it).

Q19. Which of the following is a method of feature extraction?

A. Principal Component Analysis (PCA) ✓

B. One-Hot Encoding

C. Min-Max Scaling

D. Z-normalization

I picked A: correct

Why: PCA creates new features from the originals, so it is feature extraction.

Others wrong because: One-Hot Encoding converts categories to binary columns (encoding). Min-Max Scaling and Z-normalization rescale values (preprocessing); none of them create new informative features.

Remember: PCA = extraction. Scaling/encoding = preprocessing.

Q20. What type of learning involves learning from labeled data?

A. Supervised Learning ✓

B. Unsupervised Learning

C. Reinforcement Learning

D. Semi-supervised Learning

I picked A: correct

Why: Supervised learning learns from data where each example has a label (the correct answer).

Others wrong because: Unsupervised uses no labels. Reinforcement learning learns from rewards through interaction. Semi-supervised mixes a small amount of labeled data with a lot of unlabeled data.

Remember: Labeled data = supervised.

Q21. Which of the following transformations is most appropriate for dealing with skewed data in a regression problem?

A. Logarithmic Transformation ✓

B. One-Hot Encoding

C. Polynomial Feature Expansion

D. Normalization to [0, 1] range

I picked A: correct

Why: A log transform compresses large values much more than small values. This pulls in a long right tail and makes the distribution closer to symmetric, which helps regression.

Others wrong because: One-Hot Encoding is for categorical data. Polynomial expansion adds non-linear features and does not fix skew. Min-max scaling keeps the same shape (still skewed), only changing the range.

Remember: Skewed data -> log transform.

Q22. Which of the following methods is not typically used for feature scaling in machine learning preprocessing?

A. Min-Max Normalization

B. Z-score Standardization

C. One-Hot Encoding ✓

D. Robust Scaling

I picked C: correct

Why: One-Hot Encoding turns categories into binary columns, which is encoding, not scaling.

Others wrong because: Min-Max, Z-score and Robust Scaling all rescale numeric features.

Remember: One-hot = encoding, not scaling.

Q23. Which of the following statements about normalization is correct?

A. It always results in a normal distribution

B. It scales features to have zero mean and unit variance

C. It is only applicable to numerical data ✓

D. It increases the range of feature values

I picked C: correct

Why: Normalization works by doing arithmetic on feature values (subtracting min, dividing by range), so it only makes sense for numerical data. Categorical data must be encoded first.

Others wrong because: A is false: scaling never changes the shape of the distribution into a bell curve. D is false: normalization usually shrinks the range. B is the definition of STANDARDIZATION (z-score), not min-max normalization.

Remember: Normalization = numeric only; it does not make data Gaussian.

Note: in this exam, 'Z-score normalization' (Q7) is treated as normalization, so the course key might favor B. C is correct under the strict definition.

Q24. Which of the following is NOT a feature representation of data?

A. Bag of Words

B. k (as in k-nearest neighbor algorithm) ✓

C. Number of Hull pixels

D. Inverse Document Frequency

I picked B: correct

Why: A feature representation describes the data itself. k in KNN is a hyperparameter, a setting chosen by the programmer that controls the algorithm, not a property of the data.

Others wrong because: Bag of Words and Inverse Document Frequency (IDF) are text features. Number of Hull pixels is an image/shape-based numeric feature.

Remember: k in KNN = hyperparameter, not a feature.

Q25. When using matrix multiplication to change data dimensions, which operation can be used to reduce the number of features?

A. Transpose

B. Inverse

C. Dot product ✓

D. Element-wise multiplication

I picked C: correct

Why: Multiplying a data matrix X (n x d) by a matrix W (d x k) with $k < d$ gives an n x k matrix, which has fewer features. This is the matrix (dot) product, and it is how PCA projections work.

Others wrong because: Transpose only flips rows and columns. Inverse is only defined for square invertible matrices. Element-wise multiplication keeps the same shape.

Remember: Dot product changes the number of columns.

Q26. Which of the following is FALSE about data augmentation?

A. It helps extract more feature representations from the dataset

B. It adds more data for training than is available in the training set

C. It cannot be performed on data other than images. ✓

D. All of the above

I picked C: correct

Why: Augmentation is also used for text (synonym replacement, back-translation), audio (noise, pitch shift) and tabular/time-series data. So 'only images' is false.

Others wrong because: A and B are true statements about augmentation. D is wrong because not all of A, B and C are false.

Remember: Augmentation is not only for images.

Q27. The main objective of a classification algorithm in supervised learning is to _____.

A. predict continuous values

B. assign input data to predefined categories or classes ✓

C. determine the optimal number of clusters

D. identify patterns in unlabeled data

I picked B: correct

Why: Classification maps each input to one of a set of known categories (spam / not spam, digit 0-9).

Others wrong because: A is regression. C and D are unsupervised tasks (clustering).

Remember: Classification = pick a class.

Q28. What is the main purpose of cross-validation in machine learning?

A. To increase the computational efficiency of model training

B. To estimate the model's performance on unseen data and prevent overfitting ✓

C. To automatically select the best machine learning algorithm for a given dataset

D. To generate additional training data through data augmentation techniques

I picked B: correct

Why: Cross-validation gives a reliable estimate of performance on unseen data and helps you notice and guard against overfitting when comparing settings.

Others wrong because: A: it actually costs more computation. C: it helps compare algorithms, but it does not choose one automatically. D: that is augmentation.

Remember: CV -> trustworthy performance estimate.

Note: Q18 and Q28 are two versions of the same idea. Here the option wording includes 'estimate ... on unseen data', which is the best answer.

Q29. What is the main goal of normalization in data preprocessing?

A. To remove outliers

B. To scale features to a range ✓

C. To encode categorical variables

D. To reduce dimensionality

I picked B: correct

Why: Normalization rescales numeric features to a common range (commonly 0 to 1) so no feature dominates because of its units.

Others wrong because: It does not remove outliers, encode categories or reduce dimensions.

Remember: Normalization = scale to a range.

Q30. What is the main goal of feature extraction?

A. To increase the dimensionality of the feature space

B. To reduce the number of features ✓

C. To slow down the learning algorithm

D. To increase the predictive accuracy

I picked B: correct

Why: Feature extraction transforms the original features into a smaller set of new features that keep most of the useful information.

Others wrong because: A is opposite to the goal. C is a side effect to avoid. D may happen but is not guaranteed; the main goal is to reduce features.

Remember: Extraction = reduce the number of features.

Q31. Which of the following is an example of supervised learning?

A. K-Means Clustering

B. Linear Regression ✓

C. DBSCAN Clustering

D. Principal Component Analysis

I picked B: correct

Why: Linear Regression learns from labeled (input, target) pairs, so it is supervised.

Others wrong because: K-Means and DBSCAN are clustering (unsupervised). PCA is an unsupervised dimensionality reduction technique.

Remember: Regression and classification = supervised.

Q32. What is the purpose of a validation set in supervised learning?

A. Evaluate model accuracy

B. Train the model

C. Provide additional features

D. Fine-tune hyperparameters and prevent overfitting ✓

I picked D: correct

Why: The validation set is held out from training and used to compare settings (hyperparameters), choose between models, and spot overfitting (e.g., early stopping).

Others wrong because: A is the job of the test set (final evaluation). B is the job of the training set. C is unrelated.

Remember: Validation = tune hyperparameters.

Q33. Which of the following is not an accuracy metric commonly used in machine learning?

A. Mean Absolute Error (MAE)

B. Root Mean Squared Error (RMSE) ✓

C. F1 Score

D. Precision

I picked B: correct

Why: MAE and RMSE are ERROR metrics for regression (lower is better), while F1 and Precision are classification metrics. Strictly, neither MAE nor RMSE is an 'accuracy' metric, so the question does not single out one option.

Others wrong because: All four options are common metrics. The wording 'accuracy metric' does not clearly separate RMSE from MAE.

Remember: MAE/RMSE = regression errors. Precision/F1 = classification.

Note: the question is ambiguous. RMSE (selected) and MAE are equally defensible, so the intended key cannot be confirmed.

Q34. What is the primary drawback of using a large K value in KNN classification?

A. High computational complexity

B. Overfitting

C. Underfitting ✓

D. Low accuracy for predictions

I picked C: correct

Why: With a large K, each prediction averages over many neighbors, so the decision boundary becomes too smooth and misses local patterns. That is underfitting (high bias). A very small K (e.g., 1) is what causes overfitting.

Others wrong because: B is the problem of a small K. A is a cost that grows slightly with K but is not the main drawback. D is vague; underfitting is the precise cause.

Remember: Large K = underfit. Small K = overfit.

Q35. What is the main benefit of feature extraction?

A. Reduces dimensionality of feature space ✓

B. Speeds up the learning algorithm

C. Improves predictive accuracy

D. None of the above

I picked A: correct

Why: The defining benefit of feature extraction is a lower-dimensional feature space. Speed or accuracy gains are possible consequences, but are not guaranteed.

Others wrong because: B and C are side-effects that depend on the data and model. D is wrong because A is correct.

Remember: Extraction -> lower dimensionality.

Q36. When using data augmentation with rotation, which of the following is FALSE?

A. It always preserves pixel information ✓

B. It preserves all information for a 90degree rotation

C. It sometimes changes the aspect ratio of the images

D. Center of rotation is not alway image center

I picked A: correct

Why: Rotating by an arbitrary angle needs interpolation and either crops the corners or pads empty areas, so some pixel information is lost or invented. 'Always preserves pixel information' is therefore false.

Others wrong because: B is true: a 90-degree rotation only rearranges pixels. C is true: rotating a non-square image by 90 degrees swaps width and height, and other angles can need a larger canvas. D is true: the rotation center can be chosen.

Remember: Only 90-degree multiples are lossless.

Q37. Which of the following techniques would perform better for reducing the dimensions of a data set?

A. Removing columns that have high variance in data

B. Removing columns that have too many missing values ✓

C. Removing columns with dissimilar data trends

D. None of the above

I picked B: correct

Why: A column with too many missing values carries little usable information, so dropping it is a standard way to reduce dimensions (the 'missing value ratio' technique).

Others wrong because: A is backwards: high-variance columns are the informative ones, while LOW-variance columns are dropped. C is not a recognized technique. D is wrong because B is valid.

Remember: Drop high-missing and low-variance columns, not high-variance ones.

Q38. Let C be a truly random classifier. Let N be the size of the testing data set X (sufficiently large). We find that C gives an accuracy of ~25.2% after multiple trials. What can be said about the number of classes P in X?

A. There'll approximately be 4 classes. ✓

B. It'll depend on N

C. It depends on how P is distributed in N

D. P >= 5

I picked A: correct

Why: A truly random classifier picks each of the P classes with probability 1/P, so its expected accuracy is 1/P regardless of N or how the classes are distributed. Here 1/P = 0.252, so P = 1/0.252 = 3.97, which is about 4.

Others wrong because: B and C are wrong because the expected accuracy 1/P does not depend on N (when large) or on the class distribution. D is wrong: P >= 5 would give at most 20% accuracy.

Remember: Random accuracy 25% -> 4 classes.

Q39. Machine learning differs from statistical techniques in that machine learning methods _____.

A. have trouble with large-sized datasets.

B. are better able to deal with missing and noisy data. ✓

C. typically assume an underlying distribution for the data

D. are not able to explain their behaviour

I picked B: correct

Why: Classical statistics often assumes data follows a known distribution and works best on clean data. Machine learning methods are generally more flexible and robust with messy, noisy, incomplete real-world data.

Others wrong because: A is false (ML scales to big data). C describes statistical techniques, not ML. D is not a universal property (many ML models are interpretable).

Remember: ML handles noisy/missing data better; statistics assumes distributions.

Q40. Hyperparameters can be set based on:

A. Backpropagation

B. Gradient Descent

C. Learnt Parameters

D. Random Trials ✓

I picked D: correct

Why: Hyperparameters (learning rate, K, depth, number of layers) are not learned by training. They are chosen by trying values, for example with random search or grid search evaluated on a validation set.

Others wrong because: Backpropagation and gradient descent are the algorithms that learn the model's parameters (weights). 'Learnt parameters' are the outputs of training, not what sets hyperparameters.

Remember: Hyperparameters -> tuned by trials (random/grid search).

Q41. Let us assume we have 2 features X and Y whose ranges are [0,100] and [0,25] respectively. X and Y can be rescaled using a transformation matrix T.

A. False, as we also need to add a vector term

B. True, a 2D matrix can perfectly rescale these 2 features ✓

I picked B: correct

Why: Rescaling is multiplying each feature by a constant. A diagonal matrix T = diag(1/100, 1/25) maps both ranges to [0, 1]: [x', y'] = [x/100, y/25]. Since both ranges start at 0, no shift (translation) is needed, so a plain matrix is enough.

Others wrong because: A would be correct only if the ranges did NOT start at 0 (e.g., [20, 100]); then you must subtract an offset, which needs an added vector term.

Remember: Pure scaling from 0 -> matrix only. Shifted ranges -> matrix + vector.

Q42. What does normalization typically do to numerical data?

A. Removes outliers

B. Scales the data to a range ✓

C. Makes the data categorical

D. Increase/decrease the number of features

I picked B: correct

Why: Normalization rescales numeric values into a common range such as [0, 1].

Others wrong because: It does not remove outliers, convert data to categories or change the number of features.

Remember: Normalization = values into a range.

Q43. Which of the following normalization techniques is best suited for datasets with outliers?

A. Min-Max normalization

B. Z-score normalization

C. Decimal scaling normalization

D. Robust scaling normalization ✓

I picked D: correct

Why: Robust scaling uses the median and the interquartile range: x' = (x - median) / IQR. These statistics are barely affected by extreme values, so outliers do not distort the scale.

Others wrong because: Min-max depends on the min and max, which outliers set. Z-score depends on the mean and standard deviation, both pulled by outliers. Decimal scaling divides by a power of 10 based on the max, so it is also outlier-sensitive.

Remember: Outliers -> robust scaling (median and IQR).

Q44. What is the purpose of feature scaling in machine learning?

A. To remove outliers from the data

B. To standardize the range of features ✓

C. To increase the complexity of models

D. To decrease the dimensionality of features

I picked B: correct

Why: Feature scaling puts features on comparable ranges so that distance-based and gradient-based algorithms treat them fairly and train faster.

Others wrong because: Scaling does not remove outliers, change model complexity or reduce dimensions.

Remember: Scaling = comparable ranges.

Q45. Which of the following best describes overfitting?

A. A set of data is used to discover the potentially predictive relationship

B. When a statistical model fits the training data too closely, capturing random noise or error in the data instead of the underlying patterns ✓

C. A phenomenon that occurs during the process of learning from the data

D. None of the above

I picked B: correct

Why: Overfitting means the model fits the training data so closely that it captures random noise instead of the true pattern, so it fails on new data.

Others wrong because: A describes a training set. C is too vague and could describe any part of learning. D is wrong because B is correct.

Remember: Overfit = fits noise in the training data.

Q46. In the context of matrix multiplication for changing data dimensions, what is the primary requirement for two matrices to be multiplicable?

A. Both matrices must be square

B. The number of columns in the first matrix must equal the number of rows in the second matrix ✓

C. The number of rows in both matrices must be equal

D. The matrices must have the same dimensions

I picked B: correct

Why: To multiply A (m x n) by B (p x q) you need n = p (inner dimensions match). The result has shape m x q.

Others wrong because: A, C and D are not required. Matrices of different shapes (e.g., 3x2 times 2x5) multiply fine.

Remember: Inner dimensions must match.

Q47. What is the main purpose of normalization in machine learning?

A. To increase the range of feature values

B. To decrease the range of feature values

C. To standardize the range of feature values ✓

D. None of the Above

I picked C: correct

Why: Normalization brings features to a standard, comparable range. It is about standardizing the range, not simply enlarging or shrinking it.

Others wrong because: A and B each describe only one direction of change; normalization can enlarge or shrink depending on the data. D is wrong since C is correct.

Remember: Normalization = standard, common range.

Q48. What is FALSE about linear transformations in the context of machine learning?

A. It is a kind of data augmentation

B. It helps project existing feature vectors into a different feature representation

C. Distance between points in the original subspace and transformed subspace changes.

D. For N set of features, we will EXACTLY have a NxN sized transformation matrix. ✓

I picked D: correct

Why: A transformation matrix does not have to be N x N. Mapping N features to M features uses an N x M (or M x N) matrix, e.g., PCA reduces N features to M < N. So 'EXACTLY N x N' is false.

Others wrong because: A is true: rotations, scaling and shearing are linear transformations used to augment data. B is true. C is true in general: scaling or shearing changes distances (only rotations/reflections preserve them).

Remember: Transformation matrices need not be square.

Q49. What is the key difference between supervised and unsupervised learning?

A. Supervised learning requires labeled data, while unsupervised learning does not. ✓

B. Supervised learning predicts labels, while unsupervised learning discovers patterns.

C. Supervised learning is used for classification, while unsupervised learning is used for regression.

D. Supervised learning is always more accurate than unsupervised learning.

I picked A: correct

Why: The defining difference is the data: supervised learning needs labeled examples, unsupervised learning works on unlabeled data.

Others wrong because: C is wrong because supervised learning covers both classification and regression, and unsupervised learning is for clustering and similar tasks. D is wrong because accuracy depends on the problem. B is also true as a description of the goal, but A is the more fundamental 'key difference' (the presence of labels).

Remember: Key difference = labeled vs unlabeled data.

Note: B is also a true statement; A is the standard textbook answer.

Q50. How many parameters do Linear Regression model have?

A. 0

B. 1

C. 2

D. >1 ✓

I picked D: correct

Why: A linear regression model y = w1x1 + ... + wdxd + b has d weights plus 1 bias, i.e., d + 1 parameters. With at least one feature plus the intercept that is 2 or more, so '>1' is true in general.

Others wrong because: 0 and 1 are too few for a normal model with an intercept. 2 is true only for the special case of exactly one input feature.

Remember: Linear regression parameters = d + 1, so more than 1 in general.
